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Real-Time Microservices Based Environmental Sensors System for Hazmat Transportation Networks Monitoring

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Abstract

The transportation of dangerous goods represents one of the most critical risks particularly in urban areas. For example, every day large quantities of hazardous substances are transported by roads or railways inside areas with high density of population. Thus, to ensure urban and public safety the first challenge that we tackle is to find an environmental information system, which is able to provide risk management and monitoring. Moreover, it can capture urban transportation dynamics and offer specific services, visualization, analyses and evaluation of the hazmat risk in various high-risk zones. In addition, due to wireless technologies and real time intelligent sensors, the risk of dangerous goods can be carried out in real time. In this work, we adopt a microservices based architecture to have a real-time environmental sensors system that have highly scalable applications on cloud environment. By definition, this architecture consists of a set of loosely coupled and independently deployable services. In addition, the proposed system is conducted with a smart data collector, visualization abilities and a variety of distributed sensors in order to enhance the hazmat transportation network monitoring. Finally, we show how to integrate efficiently into the system prototype the hazmat routing container embedded microservice, and illustrate the orchestration with other microservices to improve this system with hazmat routing capabilities using various routing algorithms particularly bi-directional A* algorithm. This work constitutes the first step to build the Hazmat planning aid system that provides valuable data and knowledge to urban planner and decision makers

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1. Introduction

Accidents represent an event caused by the damage to a chemical plant, transport vehicle, and storage of hazardous substances. Hazardous materials or dangerous goods include explosives, gasses, flammable liquids and solids, oxidizing substances, poisonous and infectious substances, radioactive materials, corrosive substances, and hazardous

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wastes. Accidents involving the transportation of hazmat can result from different causes and lead to severe consequences characterized by fatalities, injuries, evacuation, property damage, environmental degradation and traffic disruption. Therefore, it is relevant, important and necessary to work towards identifying, predicting and minimizing the risk of incidents in the transportation of dangerous goods and their potential consequences. The transportation of dangerous goods currently confronts a large number of problems. Such as: i) Lack knowledge of the position, the nature and characteristics of the goods transported, and the risks associated with them; ii) No systematic data on incidents involving the transport of dangerous goods; iii) Late and incorrect information, concerning the application and control of rules and regulations of hazardous transportation; iv) No Continuous regulatory monitoring. All these facts emphasize the necessity to specify and prototype a real-time environmental system for hazmat transportation, which can track in real time hazmat vehicles and provides information, knowledge and services that helps decisions makers and urban planners to supervise hazmat transportation and to better reduce the risk of moving hazardous materials. This work focuses on providing as single system various cooperating services such as real-time hazmat monitoring, risk-knowledge, hazmat routing, and transportation document, in order to provide more effective and reliable transport problem solutions aimed at monitoring dangerous goods transportation in Mohammedia City. The global system is designed as a microservice based architecture as defined in (Daya, 2016). *It is an architecture style, in which large complex software applications are composed of one or more services. Microservice can be deployed independently of one another and are loosely coupled. Each of these microservices focuses on completing one task only and does that one task really well. Moreover, this type of architecture can allow building a large-scale system capable of evolving and adopting new standards and services.* The rest of the paper is organized as follows: in Section 2, we begin with highlights related work specifically within the context of hazmat transportation management and microservice-based architectures. Section 3 presents an overview of microservice architectures before describing in detail the proposed system in section 4. Sections 5 gives the main technical part of the paper and describes the architecture implementation of the prototype, in addition to the performance evaluations. We conclude the paper in Section 6 and address areas of future work.

2. Related works

The transportation of dangerous goods is one of the most complex and most safety requiring transportation technologies. Many researchers have studied transportation monitoring, tracking and vehicle routing problems. In (Boulmakoul, 1999) the authors give general specifications for a monitoring system for the transportation of hazardous materials. The paper (Planas, 2008) presents a monitoring and intervention system for the transportation of dangerous goods based on systems used in air traffic control. This system aims to provide civil security centers with real-time knowledge of the position and contents of dangerous vehicles. An open architecture for real time GPS data collection in addition to its performance and scalability tests is explained in (Laarabi, 2012). (Erkut E. T., 2007). The authors of (Yılmaz, 2016) give a literature survey about the hazmat transportation articles classified according to risk, routing and scheduling, emergency response, network design and accident analysis.

As microservice is relatively a new concept, there are a few works in the field of microservices-based architecture. Basic information about the microservice approach is provided in (Newman, 2015). With lots of examples and practical advice, this book takes a holistic view of the topics that system architects and administrators must consider when building, managing, and evolving microservice architectures. In (Familiar B., 2015), Familiar provides a working definition of microservices and contrasts this approach with traditional monolithic layered architecture. Daya in (Daya, 2016) gives a broad understanding of this increasingly popular architectural style, and provides some examples of the way to develop applications using the microservices approach.

3. Microservices-based Architecture

In general, the objective of environmental information systems is to facilitate the collection, organization, storage, analysis, and publication of information related to an interesting phenomenon (the hazmat transportation in our case). These types of systems deal with geospatial information and services. They allow the user to store, query and process environmental information and visualize it in maps, diagrams and reports. From the point of view of system theory, environmental information systems are autonomous components, which provides different functionalities in a cooperating way. The proposed system, as an environmental information system, can be viewed as a collection of

various services, integrated with each other to form a cohesive and flexible system. Providing sophisticated data storage, management and interface functionality, the hazmat monitoring is one of the core components of the proposed system. Based on GPS and GIS technology, this component gives real-time knowledge of the position and contents of vehicles. The incidents and accidents are clear indicators and should be therefore investigated to determine the hazards that played a role in a threat. Thus, it is necessary to use a historical component in addition to a data collection module. The historical data is based on accident and incident investigation and analysis in order to discover all facts pertinent to a past incident or accident, and identify opportunities for improvements meant to avoid future similar accidents. Each hazard that is identified must be evaluated and classified. This process requires a documentation and regulation component, which facilitates the recovery of characteristics and regulations of these dangerous goods. Moreover, the choice of optimal routes for hazardous substance transportation is one of the preventive measures to minimize possible risk. Therefore, it is necessary to integrate a hazmat routing component that can provide optimal road of hazardous materials where optimality in this case means a minimum risk of transport route.

The microservices-based architecture is a best-practice approach for realizing such system requirements. It can be defined as an approach to developing a single application as a suite of small services; each running in its own process and communicating with lightweight mechanisms (Newman, 2015). The key characteristics of the microservice architecture relevant to the context of this work are described below.

- **Componentization via Services:** software componentization is a practice that break software system down into smaller easily identifiable pieces. Microservice-based architecture achieves the componentization via breaking systems down into microservices, which are capable of running independently and are responsible for its own data. In other words, the system is distributed as components exposed as a microservice running as its own process. These microservices can talk to each other but are not dependent on others. A microservice is independently replaceable and independently upgradable (Newman, 2015).
- **Decentralized Data Management:** The proposed system databases are heterogeneous; they are spatial, relational and document database. Therefore, there is no unifying schema in a central database. Microservices architectures are distributed systems with decentralised data management. This means that every service has its own, independent, storage subsystem that is isolated from other services (Namiot, 2014).
- **Technology Heterogeneity:** With a system composed of multiple, collaborating services, we can decide to use different technologies that are most suitable for each one (Abbott, 2009). Consequently, microservices architectures are usually heterogeneous with regard to operating systems, programming languages, and data stores. This use of multiple languages, frameworks and technologies for individual service, it is also known as polyglot programming approach, which gives benefits such as scalability and interoperability.
- **Evolutionary Design:** The change has historically been difficult to anticipate and expensive to retrofit; consider that over time, more capabilities can be migrated. The use of microservices-based architecture will permit to add new user experiences and new business capabilities (Familiar B. , 2015).

4. The proposed system

The objective of the proposed system is to monitor and control the transportation of hazardous materials. Such like materials are transported regularly on the roads and whenever they are involved in an accident, they could cause serious consequences for road users, surrounding communities, infrastructure and the environment. Furthermore, the term “Internet of Things (IoT)” is related to the integration of the physical world with the virtual world of the Internet. There are various physical objects, one wants to be able to track, to monitor and to interact with. The hazmat transport vehicles can be considered as one of these objects. Apply the key technology of IoT to hazmat transportation can improve system reliability and transport safety (WANG, 2009). We used the global positioning system (GPS) technology, wireless sensor networks (WSN), geographic information system (GIS), in addition to IoT technologies, to achieve real-time monitoring and tracking of hazardous materials.

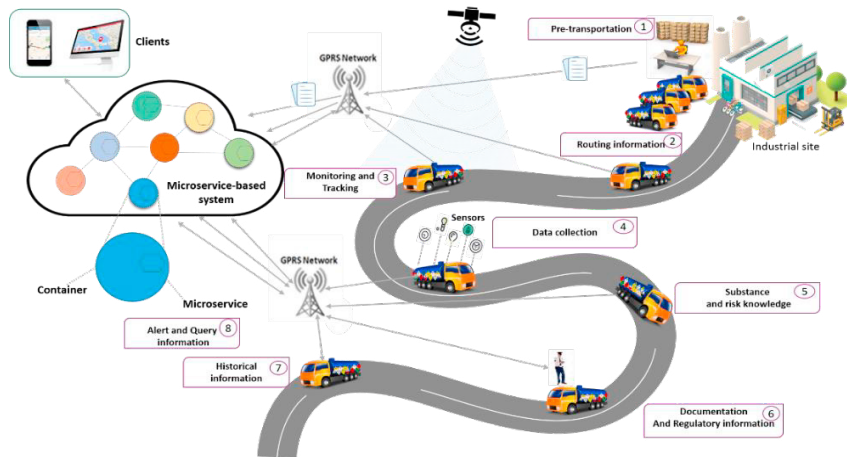


Fig. 1. Overall Open Architecture

The whole system is built based on micro-services architecture over a cloud environment and provide the following services (see figure1): 1) **Pre-transportation**: preparing and sending shipping paper, instructions in writing, and emergency response information into document management microservice. This ensures accurate communication, avoids misunderstanding and reduces waiting time. 2) **Routing**: we use a GIS database with risk costs and the pgRouting extension, in view of finding the least-risky routes for HAZMAT shipments. 3) **Monitoring and Tracking**: based on GIS web map and GPS, the system can visualize the vehicle moving on the map with all the important data and parameters right alongside it. 4) **Data collection**: this microservice communicates with in-vehicle sensors and collects sensor data (such as real-time information about goods, driver and vehicle, etc.). Once the data received, the microservice stores the row data onto the database. 5) **Substance and risk knowledge**: that permits to retrieve hazardous material description stored in the hazmat database. The hazmat information concerns the product identification, the nature of the danger, the physical and chemical properties, security instructions and transportation conditions, etc. 6) **Transport documentation**: the documentation is crucial for shipping dangerous goods by road. Therefore, providing in time the appropriate documentation for appropriate authority can dramatically reduce the time and error rate to a minimum. 7) **Historical information**: based on historical data storage, the history service can achieve the historical query and track playback, thus can improve the reliability of the system. Finally, the following sub-sections will explain in detail the main microservices providing by the proposed system. 8) **Alert and Query information**: Warning and alert notification in the case of dangerous situations can prevent accidents by allowing the protective and preventive measures to be taken.

4.1. Data collection and monitoring service

The collection of data is one of the basic requirement for any information system. It is unquestionable that hazmat data is the primary source for any kind of hazmat transportation projects and necessarily the data should be reliable and precise. Hence, the more meaningful data is obtained, the more knowledge that can be provided. Moreover, the Internet of Things (IoT) is a paradigm that allows smart devices to be connected to the Internet. Such devices can be sensors, which are able to operate and transmit data to other systems. The Wireless Sensor Networks (WSNs) are one of the most important elements in the IoT paradigm, mainly used to collect and route data from the environment to the end users. In this section, we propose a solution to collect and transfer data efficiently, using a bandwidth-efficient and energy-efficient protocol capable of providing a simple, reliable stream of data, such as Message Queue Telemetry Transport (MQTT) protocol, which is an open and lightweight publish/subscribe protocol designed specifically for machine-to-machine and mobile applications. It is optimized for communications over networks where bandwidth is at a premium or where the network connection could be intermittent (Stanford-Clark, 2013). The main feature of MQTT is publish / subscribe structure. In this architecture, a client (subscriber) needing data registers its interests with

a server (broker). The client (publisher) producing data sends the data to a broker and this broker forwards the incoming data directly to the subscribers (Eugster, 2003).

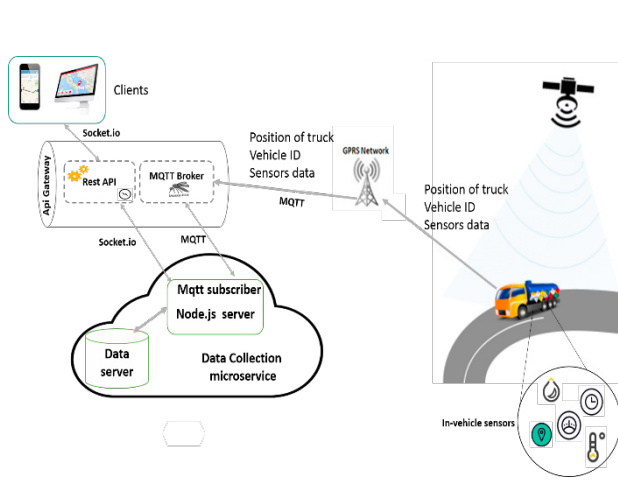


Fig. 2. Data-collection microservice architecture

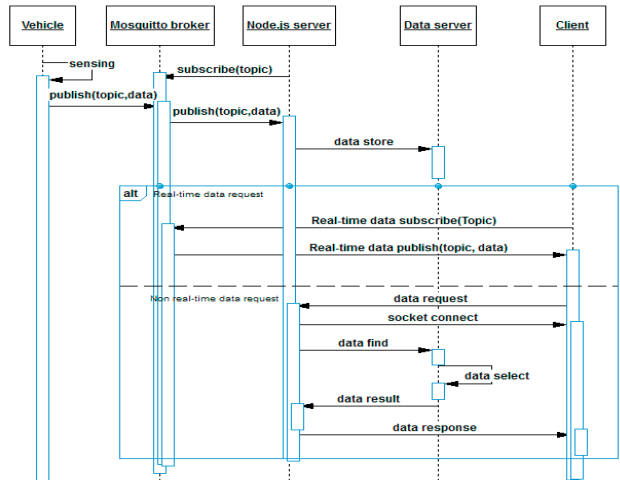


Fig. 3. Main function flow diagram of the proposal solution.

The architecture shown in figure. 2 is designed for describing our data collection solution. It is classified into : 1) **IoT devices**: represent the sensor layer in an IoT infrastructure, which is responsible for obtaining the various types static/dynamic information of the real world through various types of sensors. 2) **MQTT broker**: primarily responsible for receiving all messages, filtering them, decide who is interested in it and then sending the message to all subscribed clients. We use the open source Mosquitto server as a broker for MQTT communication. 3) **MQTT subscriber server**: When the sensing data is published from the broker, the MQTT Client (Node.js server) is configured to subscribe to these data and store them in the data server (MongoDB). 4) **Clients**: Represent the application layer in an IoT infrastructure, which exploits the collected data to provide a continuous monitoring of hazmat vehicles. The figure 3 presents the sequence diagram of our proposal data-collection solution: Starting from the vehicle, using the GPS satellite navigation system for receiving the position of the vehicle, the position and the vehicle ID are transmitted in addition to in-vehicle sensors data to the MQTT broker using MQTT protocol over GPRS network. The Node.js server first asks the Mosquitto server to subscribe to the data it wants to store in the data server. Once the data is published and forwarded, the node.js server stores it in Mongoddb database. When the client requests for non-real-time data, the Node.js server sends the requested data, after checking and extracting it from the data server. For the real-time data request, the client asks Mosquitto server to subscribe to the specific topic. Then, the Mosquitto server sends the data to the subscribing clients. A client web-mapping application based on many benefits of GIS was developed to show knowledge (non real-time data) and movements (real-time data received) of transported materials that can enable possible accident.

4.2. Risk knowledge and transport documentation service

Information such as the quantity, description of the goods, and the condition in which they were shipped are crucial for hazmat transportation management. This information gives the control authority a risk knowledge information in order to produce decisions. Using a risk knowledge database, the proposed system provides static hazmat data (substance identification, Physico-chemical properties and recommended emergency treatment), in addition to the digital form of transport documentation for facilitating its control.

4.3. Routing service

To assess the best routes for the transport of dangerous goods, a microservice based on a combination of the geographical information system (GIS) with the spatial database and the pgRouting bi-directional A* algorithm was

proposed. It aims to manage, treat, and represent geographical data related to the evaluation of the risk of transport on a road network. Two kinds of data are needed as input for the proposed service:

- Spatial data, which represents the transportation network as a collection of arcs with nodes loaded from OSM (OpenStreetMap) through QGIS (QGIS User Guide) server and stored in spatial database PostgreSQL/PostGIS.
- Attribute data, which represents the values of risk on an arc. However, these values are calculated as the product of both the probability and the consequence of the undesirable event (Erkut E. &, 1998).



Fig. 4. Architecture of routing microservice.

The figure 4 illustrates an overview of the proposed system workflow. Firstly, the clients send a routing query that contains a source and a destination to the routing server by a web-mapping application. Secondly, on the routing server side, the server connects to the PostgreSQL/PostGIS and pgRouting bi-directional A* algorithm is executed, with the risk value taken as the criteria for optimization. Finally, the routing result sends back to the clients. The client side application was built using AngularJs, leaflet and scripted using JavaScript.

5. The prototype

5.1. The architecture implementation

In this section, we describe our prototype that is capable of providing a website with a user interface allowing end-users to exploit the internal requirements, which are developed as a distributed system of lightweight, narrowly focused, independent services. The figure 5 shows the architecture implementation of the prototype.

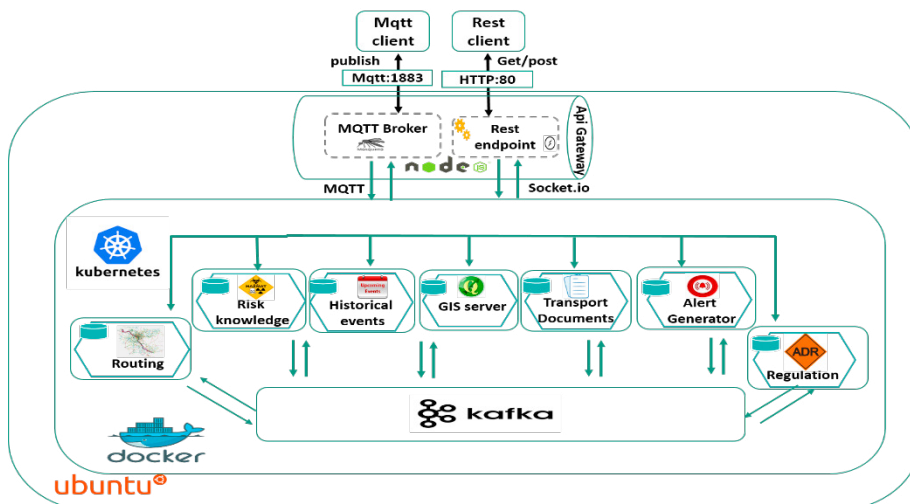


Fig. 5. The architecture implementation of the prototype

In this implementation, we use an API gateway, in order to provide a unified single entry point for external consumers that is independent of the number and composition of internal microservices. It is responsible for request routing, composition, and protocol translation. All requests from clients first go through the API Gateway. It then routes requests to the appropriate microservice (Gateway, 2016). The API Gateway feature includes two main interfaces REST and MQTT for supports REST and MQTT (using Mosquitto broker) communications. The Web front-end server is written in Node.js enhanced with Express web application framework at server side, and angular.js for the client side. Moreover, we use an event-based bi-directional communication layer for real-time web applications namely socket.io. Microservice architecture requires an adequate ecosystem that permits to obtain all the desired benefits. Indeed, microservices architecture increases the complexity of the system, which becomes a distributed system. However, multiple micro-services on the same machine running without proper isolation may affect too much the functioning and performance of each microservice. On the other hand, virtual machines for each microservice may be too heavyweight as a solution (Mozaffari, 2016). Container technology, and Docker (Docker, 2015) in particular, offer an isolated environment to run each microservice. This brings a new opportunity for building microservices. The services can be wrapped into containers with relatively small overhead, and, being virtually independent from each other. They can be managed as separate components (Familiar B. , 2015). This technology is extensible and works with other software development tools to realize a scalable build of microservices. This work employs Docker Engine, Docker Machine, Docker Compose and Kubernetes (Vohra, 2016) to realize the scalability of the Microservice Architecture. Kubernetes is a container orchestration middleware released by Google (Burns, 2016). It aims to provide an environment to schedule, deploy and manage groups of containers on a cluster of physical or virtual machines. Each container is uniquely addressable using an IP address and communicate using apache Kafka. The system was running on Ubuntu 16.04 operating system.

5.2. Evaluations of The prototype performance

The test starts with a minimum of 50 publishers, and tries to reach 750 publishers. The message rate is set to 1000 message per second per publisher. For example, 50 publishers produce 50 000 messages per second through 50 MQTT connections. The message payload size is fixed and set to 100 bytes. The time consumed for collecting messages varies between 13.07s when collecting 50 000 messages from 50 publishers and 100 s when collecting 700 000 messages from 700 publishers. The figure 6 describes the number of messages per second when connecting 50, 100, 150, 200, until 700 simultaneous publishers.

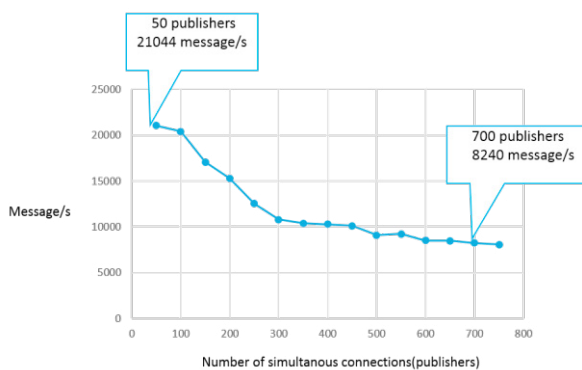


Table 1. Performance test result.

Containers	CPU (%)	Memory (MB)	Traffic
Node.js server	2.44	155.1	578.4 kbits/s
Broker server (Mosquitto)	0.90	15.7	1.3Mbits/s
Data server (Mongodb)	6	195	9.2 Kbits/s

Fig. 6. Data collection system test using 1000 messages from each publisher

The table 1, summarizes the performance test result for the node.js server, broker server, and the data sever containers, when collecting 100 000 messages from 100 publishers.

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6. CONCLUSION AND FUTURE WORKS

The main objective of this work is to provide a system for Hazmat transportation management based on microservices oriented architecture. A data collection solution is proposed to facilitate data transfer between vehicles and IT infrastructure. In addition, the routing microservices is built based on the pgRouting bi-directional A* algorithm with the risk value taken as the criteria for optimization. This affords the calculation of the least-risky routes for hazmat shipments. Moreover, the system is entirely developed based on an open source software, and its open architecture makes easy the integration of various services and scenarios. In the future works, we plan to integrate more useful services; and thus to build a Hazmat planning aid system that provides the necessary tools to get emergency response in order to improve the safety of dangerous goods transportation.

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